Instrumentation Lab Session 3 Frequency Response Functions

1. Introduction

In this experiment we will combine the input transducer with the conditioning electronics to build a working sound pressure sensor. We will then characterise the components of the system in the frequency domain by measuring and analysing their **Frequency-Response Functions**.

2. Putting it all together

Build the sound pressure sensor by following Figure 2-1:

- Connect the microphone output to the pre-amp input.
- Connect the pre-amp input to ELVIS BNC1 (or banana A and B) and Tektronix CH1.
- Connect the pre-amp output to ELVIS BNC2 (or banana C and D) and Tektronix CH2.

Now provide the sound source:

- Connect FGEN to the piezo sounder.
- Set up FGEN for a sine wave at 1.5kHz, 0.3 V_{pp} and turn it on.

Perform some measurements:

- Set up Tektronix to view both CH1 and CH2 and make sure that everything works as expected. Does the amplifier amplify the signal?
- Set up ELVIS DSA to analyse the pre-amp output (i.e. amplified sensor output).
- Increase the FGEN frequency up to 10kHz in step of 250Hz (while listening to the noise emitted by the sounder) and observe:
    - The signals on the oscilloscope
    - The power spectrum on the DSA
- What happens to the sound amplitude as the frequency changes? Notice that the voltage driving the sounder is *always* 0.3V.

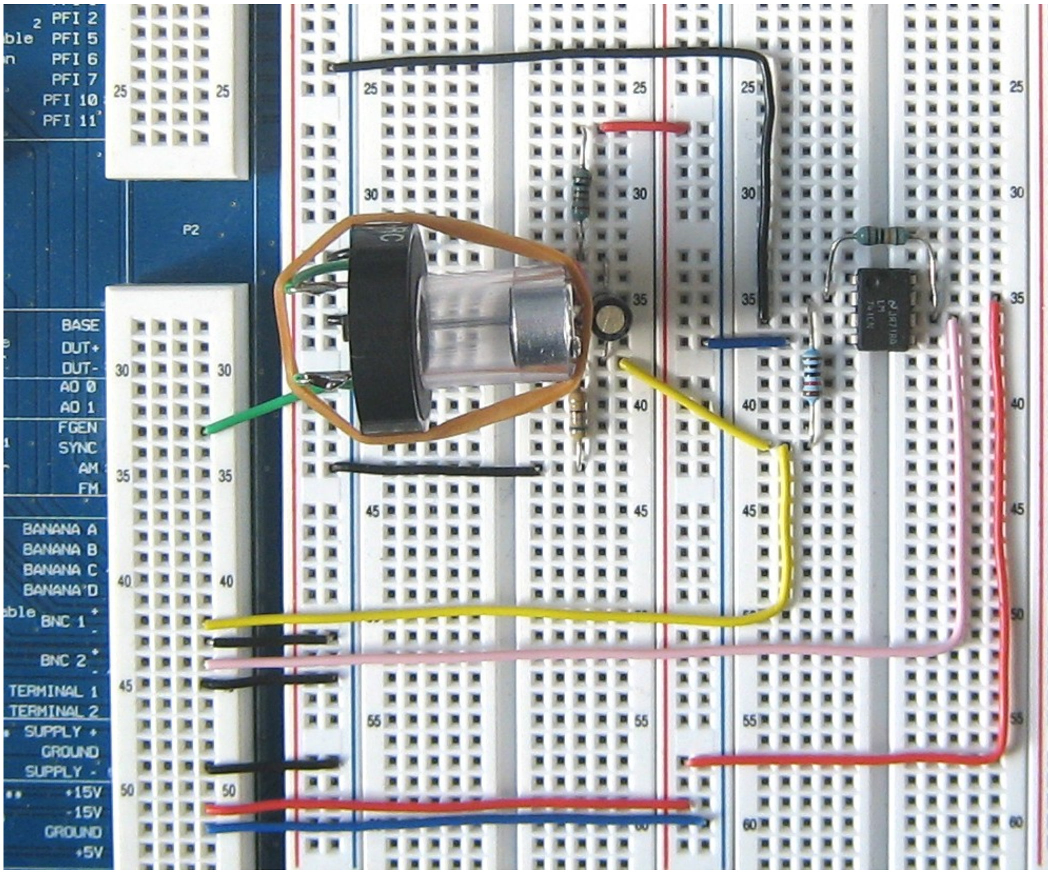


Figure 2-1, Putting it all together

3. Frequency Response

In lectures we have seen how to characterise the frequency response of a device (e.g. an amplifier or a filter) using the Bode Plot, which gives us the Gain and Phase Difference of the output, compared to the input, as a function of frequency. The Bode Plot encapsulates everything we need to know about the device in order to predict its output for any given input. We can see that this is true intuitively, since we know that any input signal can be considered as a summation of Fourier components. The Bode Plot tells us how the device modifies the gain and phase of each of the inputs frequency components. The mathematical description of the Bode Plot is the Frequency Response Function of the device, usually denoted G, where

$$G = \frac{X}{Y}$$

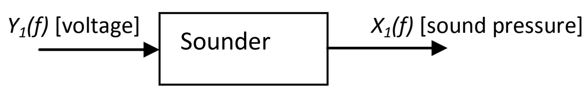
Where by convention Y is the input to the device and X is the output of the device. As stated above, we need to ensure that the input of the device contains all information about the magnitude and phase of the input signal, and hence $Y(f)$ and $X(f)$ are functions of frequency, i.e. the Fourier Transform of the input and output signals respectively. Clearly then G is also a complex function of frequency

$$G(f) = \frac{X(f)}{Y(f)}$$

Sometimes for simplicity we simply use the first form above and assume that the use of capital letters indicates that the quantities are all complex in the frequency domain.

3.1 Practical Application

The Frequency Response describes the dynamic behaviour of a device. In this experiment, we are trying to find the frequency response for the **sounder**. This is an **output transducer** which converts an electrical signal into a time-varying sound pressure output. It takes an AC voltage as an input and produces sound, i.e. air pressure changes as an output. It is an electro-mechanical system based on piezo-electric material. Its dynamic behaviour (i.e. how it behaves when being driven by signals with varying frequency) is very complicated to model. The easiest way to characterise its dynamic behaviour is to *measure* its Frequency Response:



$$G_1 = \frac{X_1(f)}{Y_1(f)}$$

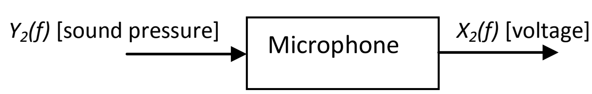
Figure 3-1, Sounder Frequency Response Function

Why would we want to do this?

It may be that we plan to use the sounder as part of an experiment where we wish to excite mechanical vibrations in a sample. As we have seen in the last lab, the sounder's output level ('volume') varies dramatically with frequency. In order to understand how much mechanical energy the sounder is putting into the sample, as a function of frequency, we need to know the frequency response function.

3.2 The Frequency Response of the System and its Component Parts

To detect the sound pressure generated by the piezo sounder we need a device that will transform sound pressure back into a measurable quantity, i.e. we need a microphone which will give us voltage. The microphone we have chosen is an **Electret capsule,** which requires some electronics to make it work.



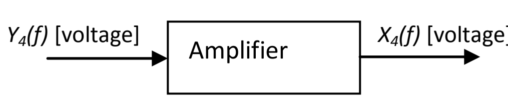
$$G_2 = \frac{X_2(f)}{Y_2(f)}$$

Figure 3-2, Microphone Frequency Response Function

$$G_3 = \frac{X_3(f)}{Y_3(f)}$$

Figure 3-3, Filter Frequency Response Function

The voltage signal produced by the microphone needs to be **conditioned** to make it suitable for our signal measurement instrument (the ELVIS digitiser). This usually means scaling the signal by means of an amplifier, filtering it to remove unwanted frequencies (either high or low or both), making it linear, etc.



$$G_4 = \frac{X_4(f)}{Y_4(f)}$$

Figure 3-4, Amplifier Frequency Response Function

Figure 3-5 gives a representation of the system you have built on the ELVIS prototyping board. There are four logical components to the system each of which has an input and an output.

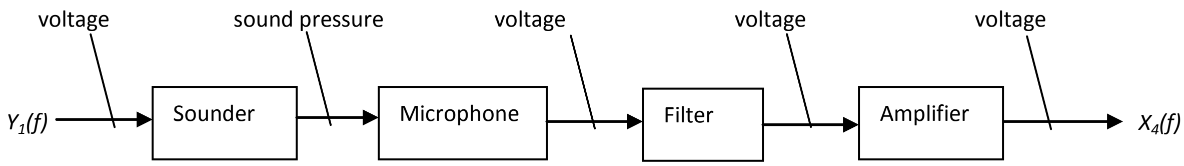


Figure 3-5 Block Diagram of the Whole System

The frequency response of the system is:

$$G = \frac{X_4(f)}{Y_1(f)} = G_1.G_2.G_3.G_4$$

The frequency response is a complex function of a real variable; as such its output can be expressed in terms of Real and Imaginary parts or, by adopting the polar representation, in terms of Magnitude (or Gain) and Phase. According to the arithmetic of the polar representation:

$$|G| = |G_1|.|G_2|.|G_3|.|G_4|$$
$$\phi = \phi_1 + \phi_2 + \phi_4 + \phi_4$$

3.3 Stimulus and Response

It is possible – at least in theory – to **measure** the frequency response of each of the four components or the whole system: all you need to do is to apply to the component under investigation a suitable known stimulus and measure its output. The stimulus must be such that it will allow you to explore the behaviour of the component in the range of frequencies of your interest (theoretically from DC Hz.) Three kinds of stimuli are used:

- ❖ Swept sine excitation: a signal containing one frequency only is applied to the system and the output measured. The frequency is then increased and the process repeated, while sweeping from DC to ∞.
- ❖ Impulse: the ideal impulse (or delta function) contains all frequencies. Applying an impulse means exciting *all* frequencies at the same time.
- ❖ White noise: white noise, by definition, contains all frequencies. Applying white noise means exciting *all* frequencies.

Obviously, when dealing with real systems there is no such thing as ∞Hz. The measurement instruments used will limit the frequency band you can explore. In addition, the user may decide that it is of no interest to study the behaviour of the system beyond a certain frequency.

The swept sine is the method employed by the **Bode Analyser** (you have used this instrument in Lab 1 to measure the frequency response of the anti aliasing filter) and it is the method usually adopted to study electronic circuits. It is the method that gives better results in terms of "quality" of the measurement since at any given time we are concentrating the energy of the input signal in one frequency only. This will maximise the signal to noise ratio (SNR). The drawback is that the measurement is slow.

The impulse technique on the other hand is very fast: you apply one impulse and you explore all the frequencies. The SNR is poor (this can be in part compensating by applying an appropriate averaging technique). The impulse technique is widely used to measure the frequency response of mechanical structures undergoing vibrations. If you want to build an analytical model of a car or a building to study the behaviour when it is subjected to vibrations (useful to study and suppress noise in the car for instance, or to study how a building will oscillate during an earthquake) you need to know its frequency response function. The stimulus to the system must be a vibration. The common way to induce a vibration with large frequency content is to hit the structure with a hammer – this constitutes an impulse when talking of vibrations and mechanical structures. You can then measure the response of the system using an accelerometer, an instrument that measures the acceleration induced by the vibrations.

Equally, white noise can be employed as a stimulus. Imagine you want to build a vibrational model of a Stradivarius to unveil the secret of its exceptional qualities and then build replicas (note that this has been done for real!). Hitting the violin with a hammer is not a viable option. The stimulus in this case can be provided by a loudspeaker playing “white noise” which will make the violin vibrate at all frequencies (in theory). You can then measure the vibration on the violin by using an accelerometer.

4. Measuring the Frequency Response

4.1 Frequency Response of the Whole System

Let’s start by measuring the frequency response of the whole system comprising sounder, microphone, filter and pre-amp.

- Connect the system’s input and output to the Bode analyser.
- Use the following settings:
  - Start freq = 10 Hz
  - Stop freq = 100 kHz
  - Steps = 30
  - Peak Amplitude = 0.8 V
- Take a measurement. Repeat until you are sure the result is repeatable.
- Record the screen data using the **Log** button on the Bode Analyser (this will save the data displayed by the analyser as text, allowing you to reconstruct the graphs later on in MS Excel for instance). Record the data for **both Linear** and **Logarithmic** mapping of the screen.
- Make a note of the resonance peak’s frequency.

The frequency response you obtained should be similar to Figure 4-1.

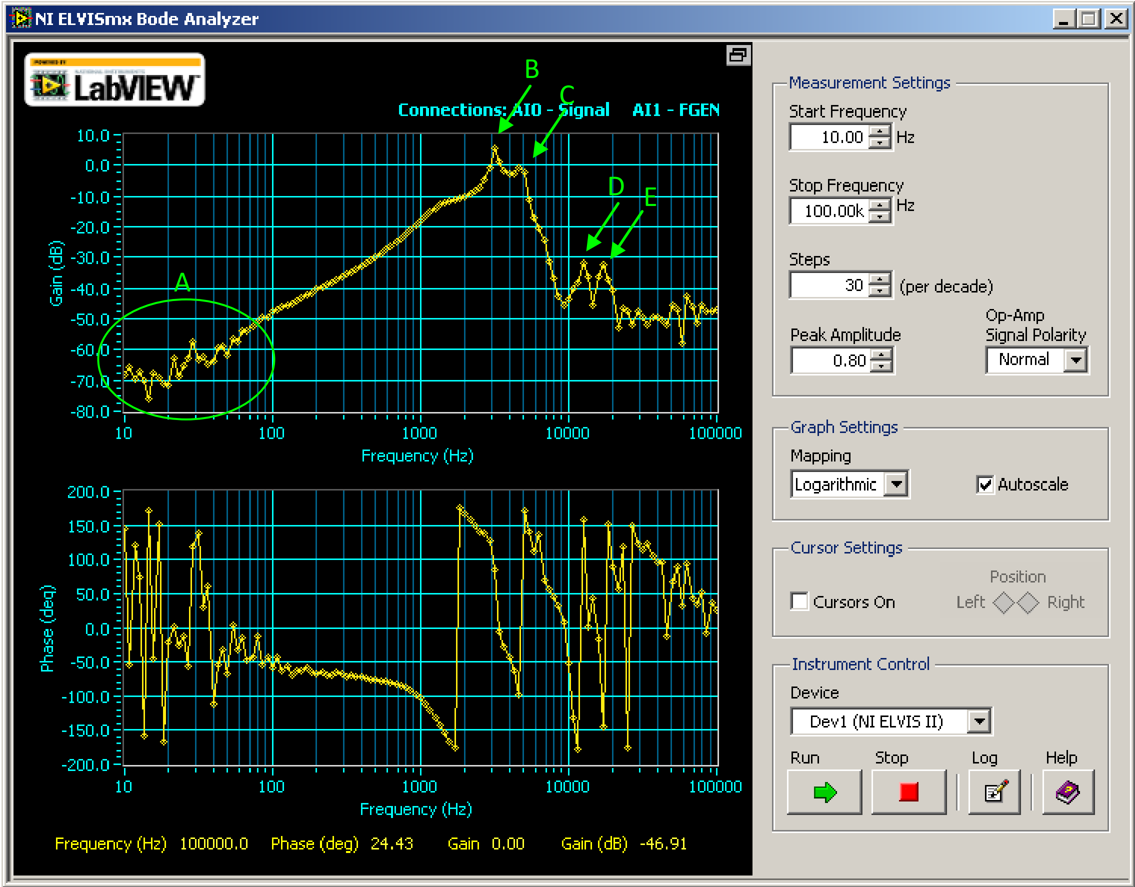


Figure 4-1 Frequency Response of the Whole System

The Bode Analyser presents the frequency response function in terms of Magnitude (Gain) and Phase.

What does the frequency response / Bode Analyser tell us in practical terms?

For a signal of frequency = 1kHz the Gain is about -18dB and the Phase is about -100deg. This means that when the system's input is a 1kHz sine wave (system's input = voltage applied to the sounder), the system's output (the pre-amp output) will be a 1kHz sine wave with amplitude attenuated (the Gain is negative) by about a factor of 8 (20 log (V_{out}/V_{in}) = -18 => V_{out}/V_{in} = 0.126). In addition, the output sine wave exhibits a Phase delay of -100deg compared to the input sine wave.

There are some features of the frequency response Gain that are worth pointing out:

1. In the low frequency region **A** the Gain doesn't show a well-defined curve but what appears to be a series of 'random' values. What is the amplitude of the output signal here in Volts? (You can get this from the plot by the method above). We may consider the output here to be too small to be physically meaningful.
2. In regions **B**, **C**, **D** and **E** the Gain curve exhibits peaks. They are **resonances**. To put it in simple terms, at those frequencies, the system seems to be able to make a much better use of the energy provided to it by giving a larger output than it does for frequencies far from the resonances (bear in mind that the amplitude of the signal provided by the Bode Analyser is 0.8V for all input frequencies)

3. The Gain curve seems not to make much sense after 25/30 kHz, for the same reason as (1) above.

   - To have a feel for what effects a resonance produce in the case of our system, use **FGEN** to scan around the peaks' frequencies and listen to the noise produced by the sounder.

The features highlighted in points 1, 2 and 3 are the result of the physical properties of the system under investigation. They are the 'sum' of the behaviour of the single components, as the frequency response just measured is the frequency response *of the whole system*. It is therefore impossible, by looking at the frequency response in Figure 4-1, to tell which of the components the resonances belong to. They may belong to the pre-amp for instance or the microphone.

- Would you buy a pre-amp with such frequency response to play your CDs? Why?
- Would you want a microphone with such frequency response?

The rest of the experiment will help you answer those questions.

Let's investigate the system a bit more and analyse the 'dynamic behaviour' of each of the components separately.

4.2 Pre-amp's Frequency Response

You have already measured qualitatively the frequency response of the pre-amp in Script 2 section 4.4. Now you will measure it using the Bode Analyser.

Firstly, let's isolate the pre-amp from the rest of the system by doing the following:

- Disconnect the filter output from the pre-amp input.
- Disconnect FGEN from sounder input.

Secondly:

- Connect FGEN to pre-amp input.
- Connect pre-amp input and output to the Bode Analyser.
- Use the following settings:
  - Start Freq = 10 Hz
  - Stop freq = 100 kHz
  - Steps = 30
  - Peak Amplitude = 0.8 V
- Take a measurement. Repeat a few times until you are sure the result is repeatable.
- Record the analyser data (**both Linear** and **Logarithmic** mapping of the screen.)
- Make a note of frequency of the peaks, if any.

The frequency response you obtained should be similar to Figure 4-2.

- Do you find in the pre-amp's Gain any of the peaks found in Figure 4-1?

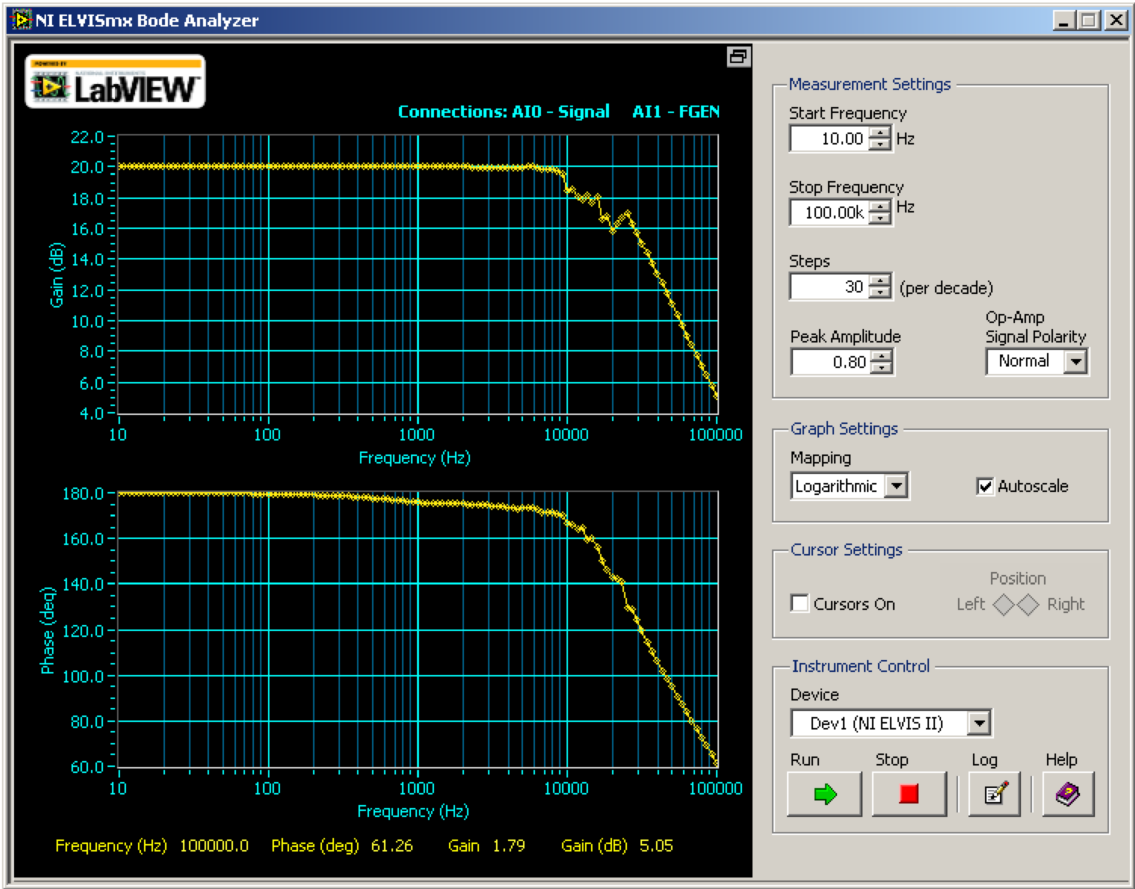


Figure 4-2 Frequency Response of the Pre-Amp

The pre-amp Gain is constant (flat curve) and equal to 20dB (i.e. a magnification of 10 times) up to about 10kHz. The practical implication of this fact is that the pre-amp will not 'alter' with respect to each other the amplitude of input signals with frequencies between DC and 10kHz; the pre-amp will only apply a gain of 20db to all of them. The same cannot be said for a frequency of 100 kHz which will be applied a gain of only about 5.

- What's the bandwidth of the pre-amp? (roughly, don't spend too much time measuring this)
- Would you buy an amplifier with this frequency response to ply your CDs? Why?

Discuss with your demonstrator if unsure.

4.3 High-pass Filter's Frequency Response

Let's analyse the behaviour of the high-pass filter. The filter is already isolated from the pre-amp. Identify the filter input and output as shown in Figure 4-3 and then:

- Disconnect the filter input from the microphone circuit
- Connect FGEN to filter input
- Connect filter input and output to the Bode Analyser.
- Use the following settings:
  - Start freq = 10 Hz
  - Stop freq = 100 kHz
  - Steps = 30
  - Peak Amplitude = 0.8 V
- Take a measurement. Repeat until you are sure the result is repeatable.
- Record the analyser screen data (**both Linear** and **Logarithmic** mapping of the screen.)

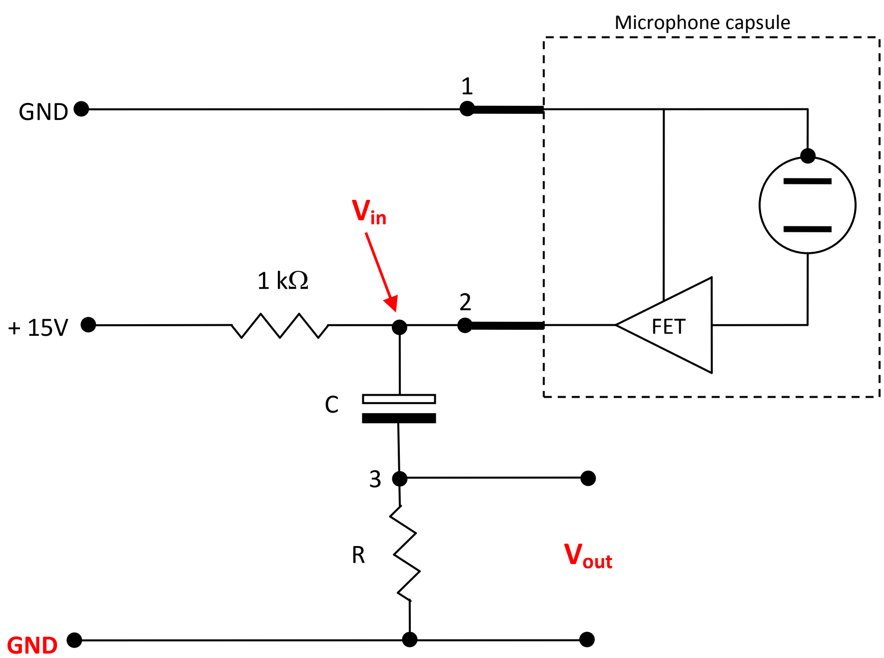


Figure 4-3 Filter Circuit Diagram

The frequency response you obtained should be similar to Figure 4-4.

- Is the measured frequency response as you would expect it to be for a high-pass filter?
- What's the filter cut-off frequency?

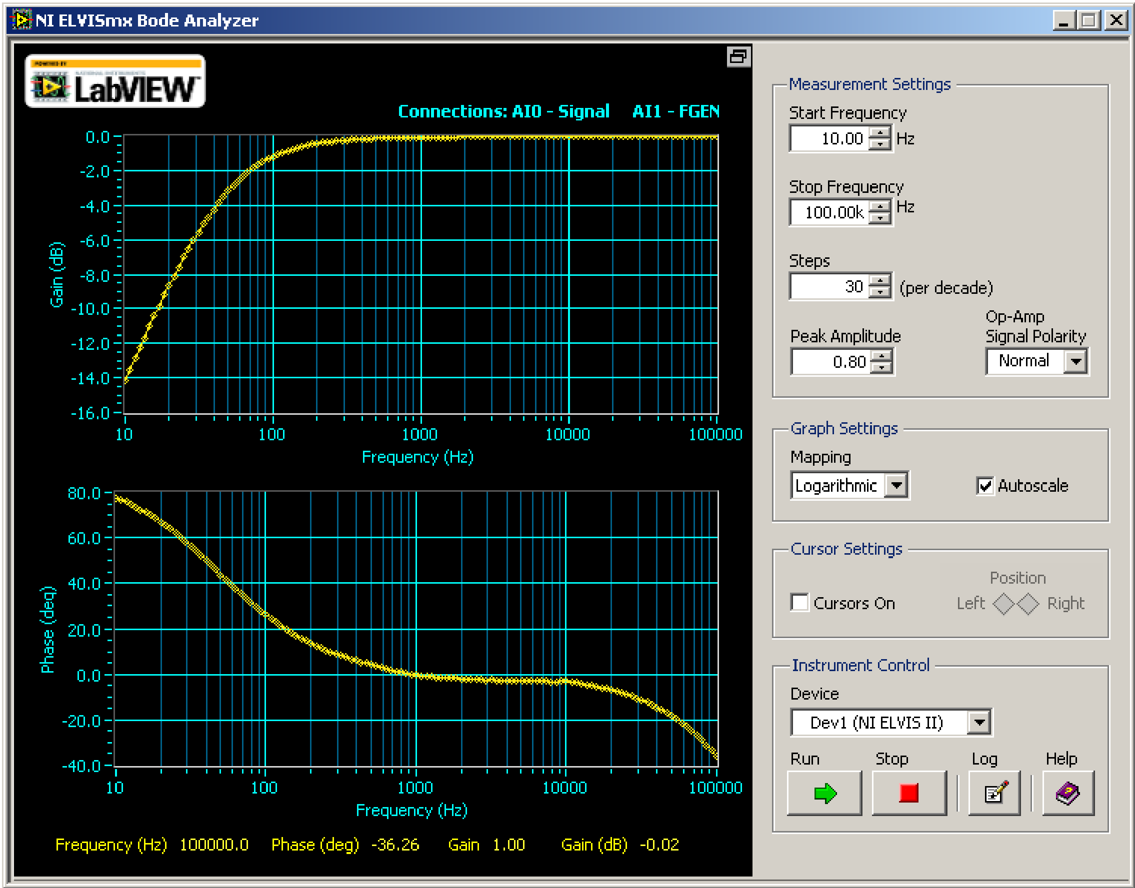


Figure 4-4 Frequency Response of the Filter

4.4 Microphone's Frequency Response

Now we want to measure the frequency response of the microphone.
The first thing to do is to identify its input and output. The output is available at the two 'legs' of the microphone. The input instead is…sound. How do I bring sound into the Bode Analyser so that we can measure the microphone's frequency function? We need a device that transforms sound into voltage. That device is… another microphone.

Measuring the frequency response of a device whose output (or input) is a quantity other than voltage is not a straightforward operation and some auxiliary components (sometimes several of them) are required. One possible set up used to measure the frequency response of our Electret microphone is shown in Figure 4-5.

The function of the reference microphone is to provide us with the 'true' value of the sound pressure as seen by the microphone under measurement. This means that the reference microphone must have exceptionally good characteristics so as to give the 'real' value of the sound pressure. The ideal reference microphone will give a voltage output proportional to the incident sound pressure at every frequency. Its frequency response should be flat.

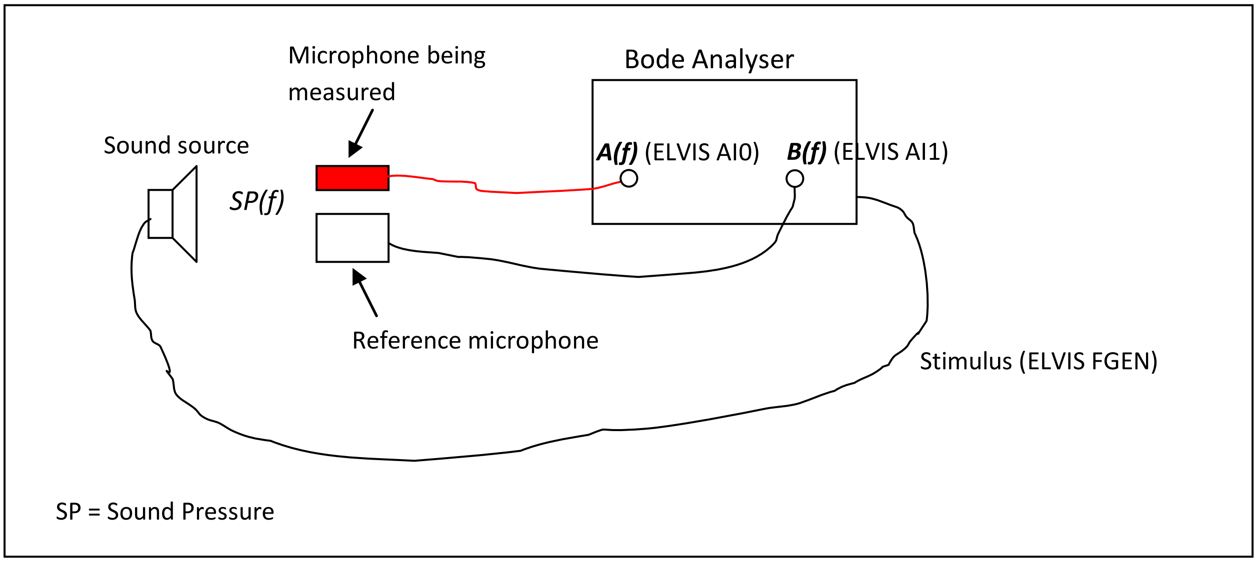


Figure 4-5 Frequency Response of the Microphone

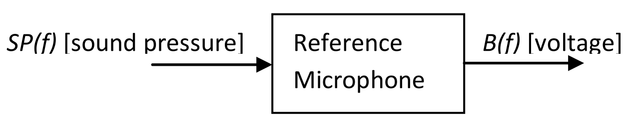
To characterise the Electret microphone we want to know:

$$G_{electret}(f) = \frac{A(f)}{SP(f)}$$

What the Bode Analyser actually measures is:

$$G_{electret}(f) = \frac{A(f)}{B(f)}$$

Considering the reference microphone's frequency respose



$$G_{ref}(f) = \frac{B(f)}{SP(f)}$$

Therefore:

$$G_{electret}(f) = \frac{A(f)}{G_{ref}(f)\,SP(f)}$$

In the appendix you'll find the data sheets of a range of microphones. See how different the specifications (and prices) can be. The microphone described as an "instrumentation" microphone is a device that was designed to be used a measurement instrument. It has a very comprehensive data sheet. Clearly, we would choose a response as flat as possible over the frequency range we are interested in, since the frequency response of the Electret (in the above equation) depends on the reference frequency response. Unfortunately we don't have one of these and therefore we will not **measure** the Electret frequency response.

Instead we will **synthesise** the Electret Gain part of the frequency response using the data provided by the microphone's data sheet (see appendix). This is of course a compromise, but it is a reasonable one. You might want to discuss this with a demonstrator. In general, experimental science requires us to make simplifications and compromises; as long as we justify these, our method is still sound. To draw the gain curve for the Electret, consider these questions:

- What shape are you expecting the Gain curve to have based on the parameters available in the data sheet?
- Can any of the curves in Figure 4-6 be used to represent the Gain curve? Why?
- What can you say about the Phase curve?
- Draw the Gain curve indicating the Electret sensitivity (i.e. the gain of the device, indicated as 'x' in Figure 4-6) and cut-off frequency(s) and discuss with a demonstrator.

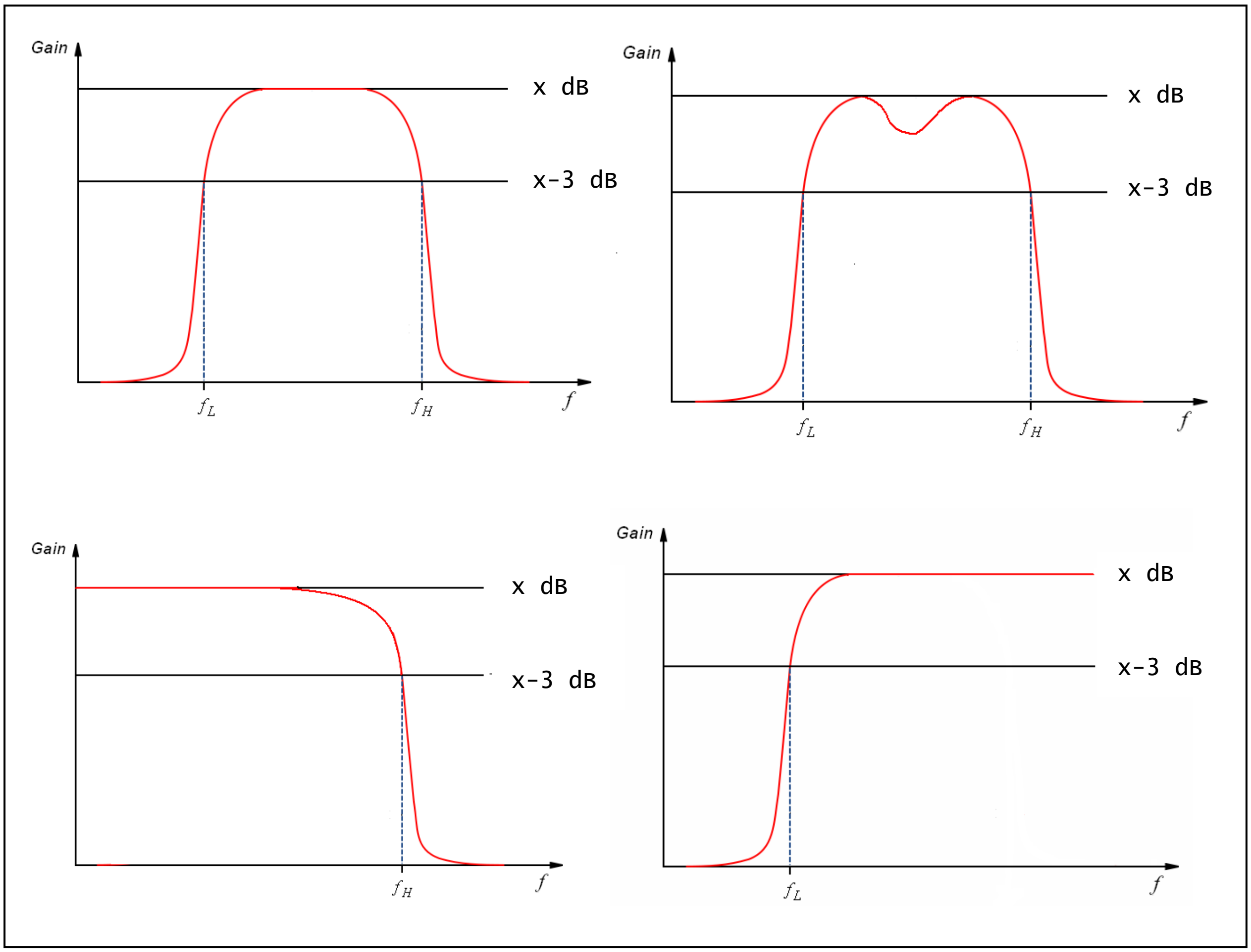


Figure 4-6 Sample Frequency Response Gain Plots for the Electret Microphone Capsule

4.5 Sounder's Frequency Response

At this stage, of the four blocks that constitute our system, only the sounder remains to be characterised.

- How would you go about measuring the frequency response of the sounder?
- What are the input and output to the sounder?
- Can you measure it using the instruments available to you?

We could consider making an indirect measurement of the Gain and Phase curves, bearing in mind how the frequency response functions combine. According to section 3.2, the sounder's response Magnitude and Phase are:

$$|G_1| = \frac{|G|}{|G_2|.|G_3|.|G_4|}$$

$$\phi_1 = \phi - \phi_2 - \phi_3 - \phi_4$$

Using the frequency response data measured and saved in Sections 4.1, 4.2 and 4.3 (use the linear values) and the frequency response synthesised in section 4.4:

- Work out in MS Excel the Magnitude of the sounder's frequency response. Use Figure 4-7 as a guideline for building the spreadsheet.
- Plot the five Magnitudes. Use Figure 4-8, Figure 4-9 and Figure 4-10 as guidelines.

	A	B	C	D	E	F
1	Freqency [Hz]	System FR Mag [V/V]	Pre-Amp FR Mag [V/V]	Filter FR Mag [V/V]	Microphone FR Mag [mV/µbar]	Sounder FR Mag [µbar/mV]
2	10.1	0.001	10.014	0.196	0.010	0.0509
3	10.8	0.001	10.013	0.210	0.043	0.0110
4	11.7	0.001	10.018	0.227	0.076	0.0057
5	12.7	0.001	10.016	0.244	0.110	0.0037
6	13.6	0.001	10.024	0.261	0.143	0.0027
7	14.7	0.001	10.016	0.281	0.176	0.0020
8	15.8	0.001	10.018	0.303	0.209	0.0016
9	17.1	0.001	10.014	0.322	0.243	0.0013
10	18.4	0.001	10.014	0.346	0.276	0.0010
11	19.9	0.001	10.020	0.370	0.309	0.0009
12	21.6	0.001	10.019	0.393	0.342	0.0007
13	23.3	0.001	10.020	0.418	0.376	0.0006
14	25.1	0.001	10.018	0.445	0.409	0.0005
15	27.2	0.001	10.015	0.472	0.442	0.0005
16	29.2	0.001	10.015	0.501	0.475	0.0004
17	31.7	0.001	10.014	0.527	0.509	0.0004
18	34.1	0.001	10.014	0.557	0.542	0.0003
19	36.9	0.001	10.016	0.584	0.575	0.0003
20	39.9	0.001	10.011	0.613	0.608	0.0003
21	43.0	0.001	10.017	0.640	0.642	0.0002
22	46.4	0.001	10.018	0.668	0.675	0.0002
23	50.1	0.001	10.015	0.695	0.708	0.0002
24	54.2	0.002	10.019	0.719	0.741	0.0004
25	58.5	0.001	10.015	0.743	0.774	0.0002
26	63.1	0.002	10.019	0.766	0.808	0.0003
27	68.2	0.002	10.018	0.787	0.841	0.0003
28	73.6	0.002	10.018	0.807	0.874	0.0003
29	79.3	0.003	10.017	0.826	0.907	0.0004
30	85.7	0.003	10.019	0.843	0.941	0.0004
31	92.6	0.003	10.013	0.860	0.974	0.0004
32	100.0	0.004	10.018	0.873	1.000	0.0005
33	108.0	0.005	10.018	0.886	1.000	0.0006
34	116.6	0.005	10.016	0.897	1.000	0.0006
35	125.9	0.005	10.018	0.908	1.000	0.0005

Point A.
Arbitrary value: choose a value which is small compared to value of Gain in the flat region of the curve.

We assume a linear curve between points A and B which then extends beyond B to meet the flat region of the FR.

The same principle is used to build the ramp at the other end of the FR between 13kHz and 100kHz.

Point B.
50Hz => -3dB point.
Gain = -63 dB = 0.708 [mV/µbar]

	A	B	C	D	E	F
81	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
82	4641.5	0.931	9.897	0.997	1.000	0.0944
83	5011.8	0.768	9.952	0.997	1.000	0.0774
84	5411.7	0.273	9.991	0.997	1.000	0.0274
85	5843.5	0.144	9.992	0.997	1.000	0.0145
86	6309.5	0.094	9.942	0.998	0.976	0.0097
87	6813.0	0.059	9.869	0.997	0.950	0.0063
88	7356.3	0.027	9.816	0.998	0.923	0.0030
89	7943.3	0.014	9.820	0.998	0.896	0.0016
90	8576.9	0.008	9.731	0.997	0.869	0.0009
91	9261.3	0.005	9.497	0.997	0.842	0.0006
92	10000.0	0.007	8.392	0.998	0.815	0.0010
93	10797.8	0.010	8.466	0.998	0.789	0.0015
94	11659.2	0.012	8.027	0.997	0.762	0.0020
95	12589.2	0.025	7.784	0.998	0.735	0.0044
96	13593.6	0.015	8.129	0.997	0.708	0.0026
97	14678.0	0.005	7.620	0.997	0.681	0.0010
98	15848.9	0.016	7.974	0.996	0.654	0.0031
99	17113.2	0.025	6.748	0.997	0.627	0.0059
100	18478.6	0.014	6.897	0.997	0.601	0.0034
101	19952.7	0.009	6.174	0.998	0.574	0.0025
102	21544.3	0.002	6.499	0.998	0.547	0.0006
103	23263.1	0.005	6.795	0.997	0.520	0.0014
104	25118.9	0.004	7.031	0.998	0.493	0.0012
105	27122.7	0.003	6.529	0.998	0.466	0.0010
106	29286.4	0.004	6.081	0.998	0.440	0.0015
107	31622.7	0.003	5.648	0.998	0.413	0.0013
108	34145.5	0.003	5.238	0.997	0.386	0.0015
109	36869.4	0.003	4.862	0.998	0.359	0.0017
110	39810.7	0.003	4.499	0.997	0.332	0.0020
111	42986.7	0.003	4.175	0.997	0.305	0.0024
112	46415.8	0.003	3.864	0.998	0.278	0.0028
113	50118.8	0.005	3.600	0.997	0.252	0.0055
114	54116.9	0.005	3.316	0.997	0.225	0.0067
115	58434.2	0.001	3.073	0.997	0.198	0.0016
116	63095.8	0.007	2.832	0.998	0.171	0.0145
117	68129.2	0.005	2.637	0.998	0.144	0.0132
118	73564.2	0.003	2.443	0.998	0.117	0.0105
119	79432.9	0.005	2.263	0.998	0.091	0.0245
120	85769.6	0.004	2.095	0.998	0.064	0.0300
121	92611.8	0.004	1.941	0.997	0.037	0.0561
122	100000.0	0.005	1.789	0.998	0.010	0.2799
123						

13kHz => -3dB point.
Gain = -63 dB = 0.708 [mV/µbar]

Figure 4-7 Calculating the Sounder Frequency Response Magnitude using a Spreadsheet

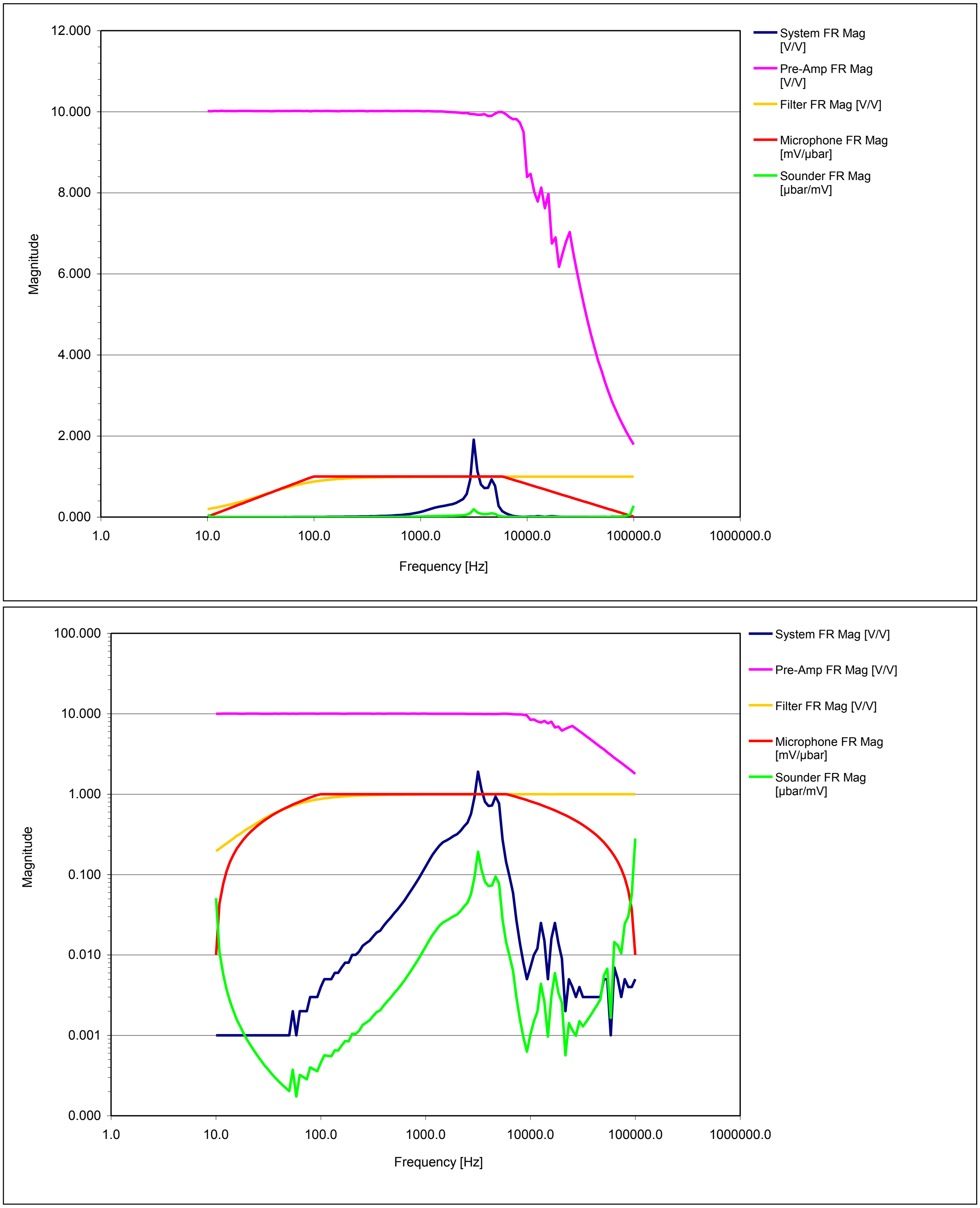


Figure 4-8 Magnitude Components from the Spreadsheet: linear and log amplitude.

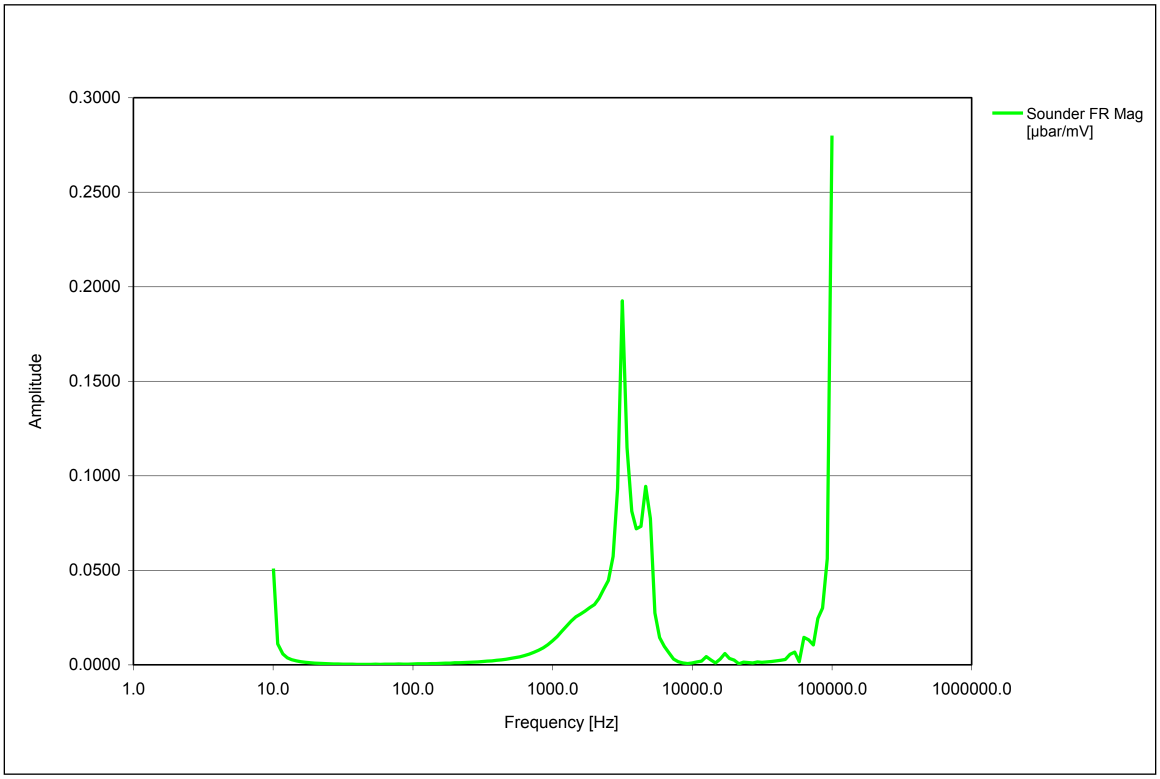


Figure 4-9 The Sounder FR Magnitude.

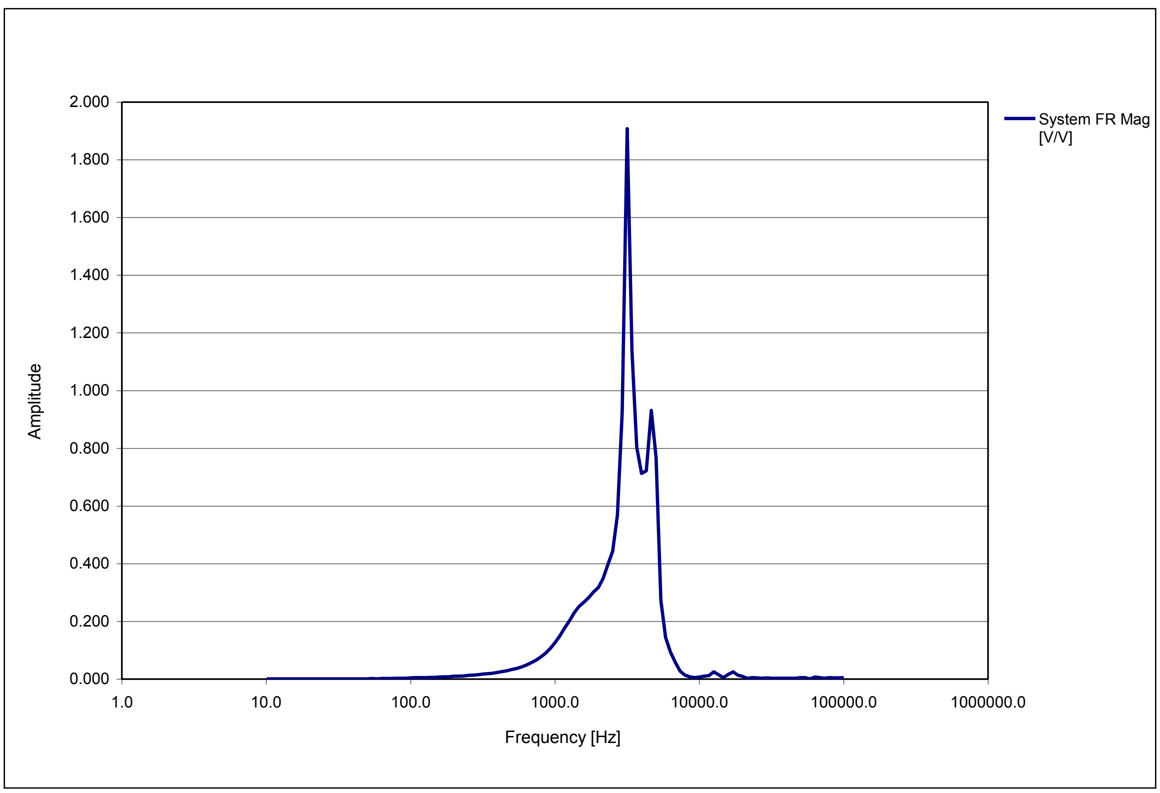


Figure 4-10 The System FR Magnitude.

Are the calculated sounder Gain values plausible? To answer this question, let's consider the greatest Gain exhibited by the sounder which is approximately 0.20 μbar/mV at about 3200Hz. This means that 1V applied to the sounder will produce 200 μbar of sound pressure **at the microphone's location** which we'll assume to be 1 cm apart from the sounder. The sound pressure varies linearly with the distance therefore at 1m from the sounder - which is where we assume the listener's ears to be – it will be 2μbar=0.2Pa (1μbar = 0.1Pa). This is true assuming that the coupling between the sounder and the microphone is not present. To put this value into context, look at the scale presented in Figure 4-11. It turns out that the sound pressure generated by the sounder at 1m is similar to that generated by a diesel freight train at 25m.

- Do you agree with this conclusion? Note that the perceived loudness of a sound is frequency dependant i.e. two noises at different frequencies exerting the same sound pressure will be perceived by the human brain as being differently loud. The human ear is most sensitive to the frequencies in the 3-4 kHz range therefore you will perceive the noise made by the sounder as "louder" than that of a train at 25m.
- Can you now say whether the calculated sounder Gain values are plausible or not?

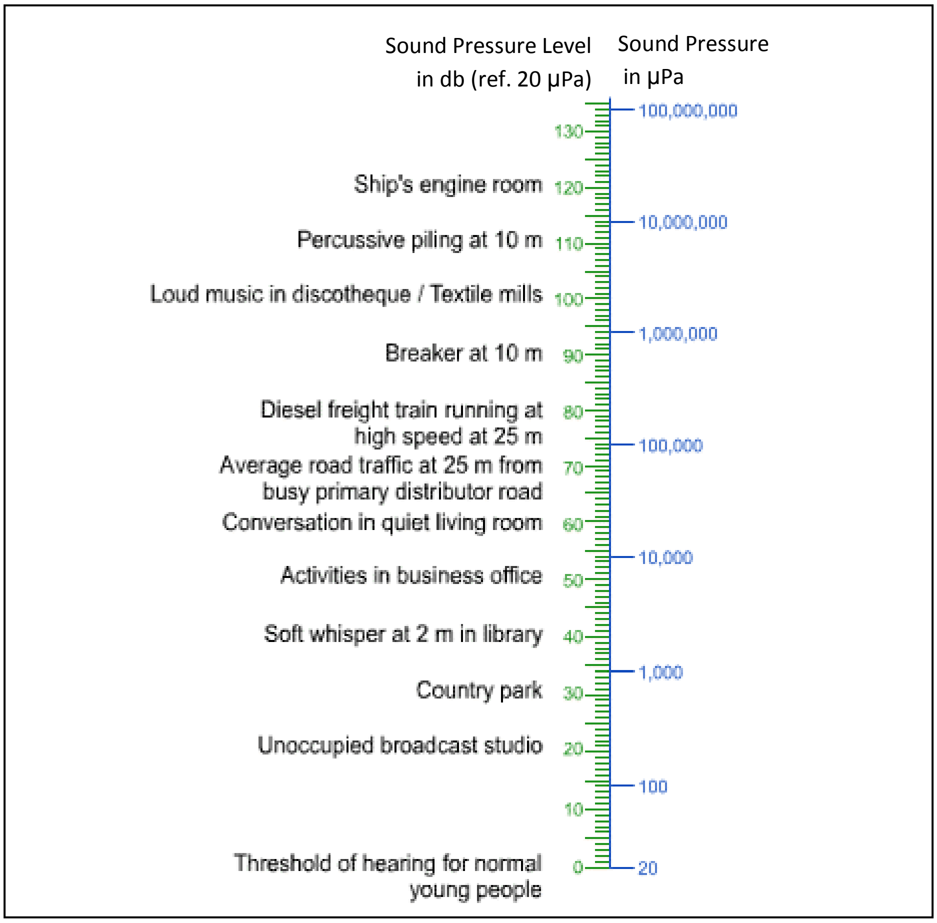


Figure 4-11: Examples of Sound Pressure vales. In addition: 0.1Pa sound pressure is typical of the sound intensity 12 inches away from a talker; 1Pa is typical at 1 inch away fro the same talker.

5. Conclusions

We have seen how it is possible to derive the frequency response of an 'unknown' component from the overall response function plus the individual functions of the 'known' components. Often, this involves making some assumptions (as in the case of the Electret where we manufactured the curve based on data from the manufacturer, not experiment). These practical methods are frequently applied in experimental physics and engineering. In the case of our sounder, we now know where its resonances lie and this would be very useful were we to use the device as a source of mechanical input to an experiment. You should be able to answer the questions below and discuss these results with a demonstrator.

- Which component(s) exhibits the resonances?
- Can you see how the microphone acts as a band-pass filter?
- Can you explain the form of the sounder frequency response for very low and very high frequencies? Are these results to be trusted?
- Can you determine the source of all the resonances you measured at the beginning of the lab (Figure 4-1)?
- Would the sounder be any use as an audio output transducer (loudspeaker) e.g. in earphones or as the speaker in a mobile phone?
- How would you go about improving the experimental setup in future to get a better result?

6. If You Have Time

6.1 Natural Sound Signals

Decouple the sounder from the microphone so that the latter can pick up the natural sound of the lab and use the dynamic signal analyser to investigate the frequency-domain characteristics of natural signals in the lab. What is the spectrum of the natural noise picked up by the device; is this as expected? You may need to boost the amplification to get a good noise spectrum. Additionally you can try making some artificial noises.

7. Appendices

7.1 Sample Microphone specifications

Instrumentation microphone

Brüel & Kjær type 4954 (price: several ££££)

Type 4954 is a 1/4-inch Prepolarized Free-field microphone and is specially designed for high-level and high-frequency measurements.

The use of a laser welded stainless steel diaphragm and a stainless steel protection grid, make this microphone capable of withstanding rough environments.

USES
- High-frequency measurements
- High-level measurements
- Cost-effective measurements

FEATURES
- Robust
- Prepolarized
- Sensitivity: 3.16 mV/Pa
- Frequency: 4 – 100000 Hz
- Dynamic Range: Up to 165 dB
- Excellent phase response
- Excellent long-term stability

Description

Prepolarized Free-field Microphone Type 4954 has been optimised for a flat, free-field response and is suited for measurements far beyond the audible range. Due to its wide frequency range, the frequency response in the audible range is typically within ±1 dB.

Carefully selected materials and advanced production techniques such as artificial aging and clean room assembly guarantee excellent long-term stability and low noise even in harsh environments.

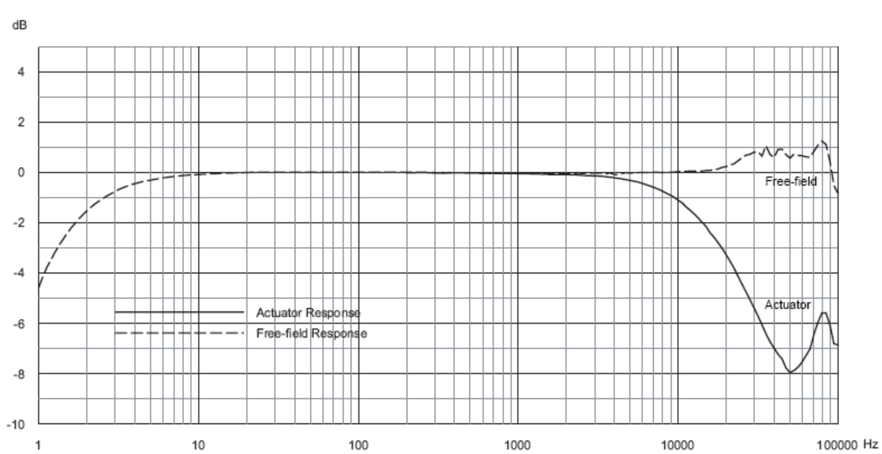
The microphone has excellent amplitude and phase matching over wide ranges of temperature and humidity.

The microphone does not need an external polarization voltage, thus enabling the use of inexpensive 2-wire system (DeltaTron®) input.

The microphone is side vented for pressure equalization.

Each microphone is individually calibrated and comes with a calibration chart and a mini CD with calibration data and information about the influence of different measurement conditions and accessories.

***Fig. 1** Frequency response without protection grid*



Specifications – 1/4-inch Prepolarized Free-field Microphone Type 4954

COMPLIANCE WITH STANDARDS
CE: Compliance with EMC Directive and Low Voltage Directive of the EU
C-tick: Compliance with the EMC requirements of Australia and New Zealand

Unless otherwise specified the data below is valid at 23°C, 101.3 kPa and 50% RH.

POLARIZATION VOLTAGE
0 V (pre-polarized)

SENSITIVITY
– 50 dB re 1 V/Pa ±3 dB
3.16 mV/Pa (re 250 Hz)

FREQUENCY RESPONSE*
Free-field response (re 250 Hz):
±2 dB, 4 Hz to 80 kHz
±3 dB, 3 Hz to 100 kHz

STANDARDS
IEC 61094 – 4 WS3F

LOWER LIMITING FREQUENCY
0.3 – 3 Hz

CARTRIDGE CAPACITANCE
5.1 pF

* Without protection grid

VENTING
Side vented

CARTRIDGE INHERENT NOISE
<35 dB SPL(A) (<1.6 µV)
<45 dB SPL (Lin, 20 Hz – 100 kHz) (< 2.52 µV)

COMBINED NOISE LEVEL (WITH PREAMPLIFIER TYPE 2670)
<40 dB SPL(A)
<50 dB SPL (Lin, 20 Hz – 100 kHz)

UPPER LIMIT OF DYNAMIC RANGE (3% DISTORTION)
>164 dB SPL (RMS)

MAXIMUM SOUND PRESSURE LEVEL
174 dB (peak)

DIAPHRAGM RESONANCE FREQUENCY
83 kHz

Environmental

OPERATING TEMPERATURE RANGE
–40 to +150°C (–40 to +302°F)

TEMPERATURE COEFFICIENT
+0.009 dB/°C @ 250 Hz (–10°C to +50°C (14 to 122°F))

PRESSURE COEFFICIENT
–0.007 dB/kPa

STORAGE TEMPERATURE
In Microphone Box: –30 to + 70°C (–22 to +158°F)
With Mini CD: –20 to +70°C (–4 to +158°F)

OPERATING HUMIDITY RANGE
0% – 100% RH without condensation

INFLUENCE OF HUMIDITY
<0.1 dB in the absence of condensation

VIBRATION SENSITIVITY
60 dB equivalent SPL for 1 ms^2 axial vibration

MAGNETIC FIELD SENSITIVITY
Equivalent to 10 dB SPL@ 80 A/m, 50 Hz field

ESTIMATED LONG-TERM STABILITY
<0.001 dB/year @ 20°C
<0.5 dB/hr @ 150°C

Dimensions

DIAMETER
7 mm (0.28″) (with grid)
6.35 mm (0.25″) (without grid)

HEIGHT
10.5mm (0.41″) (with grid)
9 mm (0.35″) (without grid)

THREAD FOR PREAMPLIFIER MOUNTING
5.7 mm – 60 UNS

Studio Microphone (high-end)

SE Electronics Gemini MK2 (price: 1100£)

Frequency Response: 20Hz-20KHz

Sensitivity: 12.6mV/Pa - 38 ± 1dB(0dB=1V/Pa 1000Hz)

Polar Pattern: Cardioid

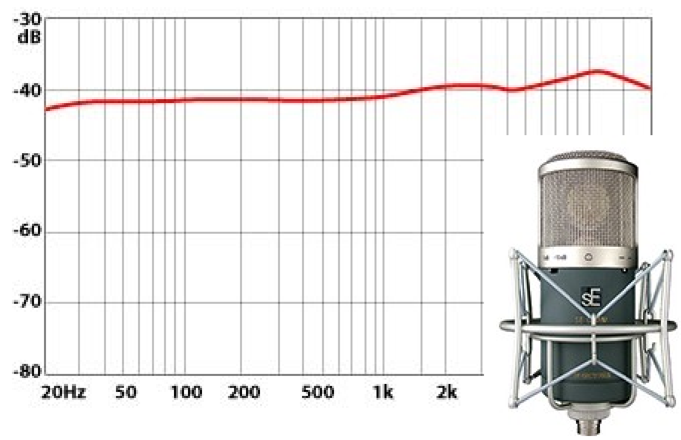
Impedance: ≤ 200 Ohms

Equivalent Noise Level: 12dB(A weighted)

Max SPL for 0.5% THD@1000Hz: 135dB

Power Requirement: External PSU included.

Connector: 3-pin



Frequency Response Amplitude

Stage microphone (mid-range)

SHURE SM58 Vocal Microphone (price: £90)

Type: Dynamic (moving coil)

Frequency Response: 50–15,000 Hz

Polar Pattern: Cardioid

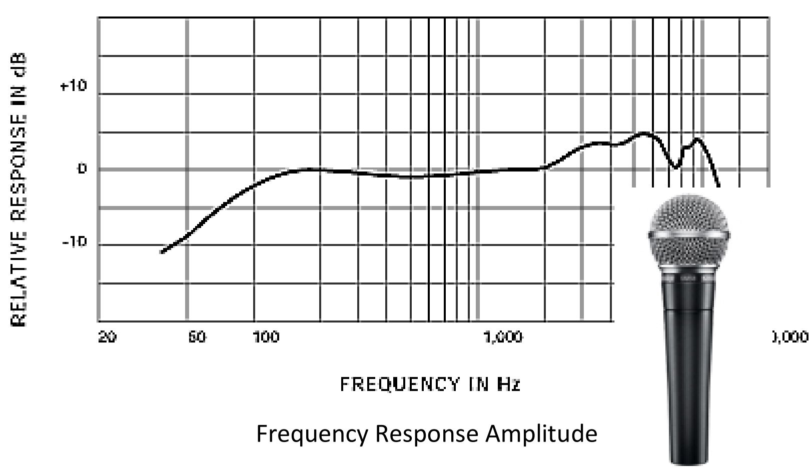
Output Impedance: EIA rated at 150Ω (300Ω actual)

Sensitivity (at 1 kHz, open circuit voltage): –54.5 dBV/Pa (1.85 mV)

Polarity: Positive pressure on diaphragm produces positive voltage on pin 2 with respect to pin 3

Weight Net: 0.298 kg (0.656 lb)

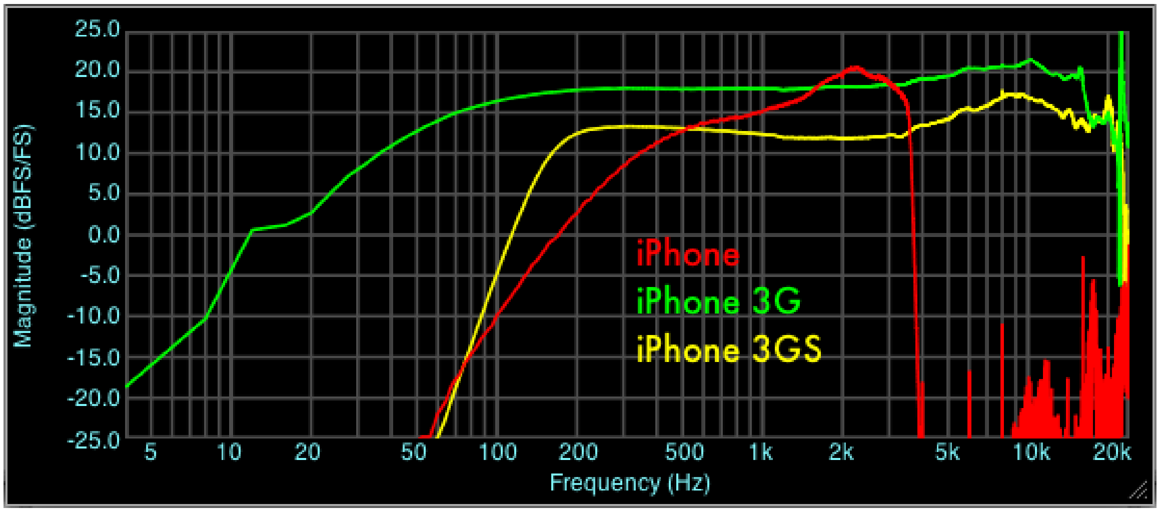
Connector: Three-pin professional audio (XLR), male



Frequency Response Amplitude

Commercial microphones (Low-end)

iPhone Microphones (price: £)



Frequency Response Amplitude (http://blog.faberacoustical.com/2009/iphone/iphone-microphone-frequency-response-comparison/)

7.2 Electret Microphone Capsule Technical Specifications (from datasheet)

Manufacturer:	unknown (cost: £)
Response:	Omni-directional
Bandwidth:	50Hz to 13kHz
Sensitivity:	-60dB ±3dB (0dB = 1V/µbar at 1kHz)
Impedance:	1kΩ maximum
S/N ratio:	>40dB
Sound pressure level:	120dB maximum
Power supply:	15V to 10Vdc
Current drain:	<4mA

7.3 Piezo-Electric Sounder Techinical Specifications (from datasheet)

Rated voltage	(V_{pp} Square Wave): 9V_{pp}
Operating voltage:	1-30Vp-p
Rated current at rated voltage:	3mA
Capacitance at 120Hz:	17000± 30%pF
Sound output at 4000Hz at 10cm at rated voltage:	>85dB
Resonant frequency:	4000± 500Hz
Operating temperature:	-20 ~ +60°C
Storage temperature:	-30 ~ +70°C
Weight:	3g